

ELECTRIC FIELDS AND ELECTRIC POTENTIALS

Laboratory Worksheet

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A. TWO RING ELECTRODE CONFIGURATION

(E_{avg} = change in potential/distance E field flows in the direction of decreasing potential)

What is the magnitude of the electric field between the 3 and 5 V equipotentials (on centerline)? Indicate the direction on the graph paper.

$$2V/0.07m = 28.6N/C$$

What is the magnitude of the electric field between the 1 and 3 V equipotentials (on centerline)? Indicate the direction on the graph paper.

$$2V/0.05m = 20 N/C$$

What is the magnitude of the electric field at coordinate (13,7)? Indicate the direction on the graph paper.

$$4V/1 = 4N/C$$

How can you describe the shape of the equipotentials at different regions of this electrode arrangement?

The shape of the equipotential lines at different regions of the electrode arrangement follows a circular path since electric field lines would be perpendicular to the equipotential levels. Half of the lines face one direction while the other half faces the other direction because the two rings' potential interact with each other and bounce off each other. The potential at 4V forms a line in the center.

Where is the E-field strongest (along the centerline)?

In the middle of the center line the electric field is the strongest because the spacing between the equipotential lines is the smallest

Where is the E-field weakest (along the centerline)?

The E-field will be weakest further away from the middle so when the distance of the middle to the equipotential lines are the furthest away.

What are the potential and the E-field inside the circular electrodes?

On the left side, the potential is 10 volts, while on the other side it is zero volts, the E field for both would be zero since conductors have a E field of zero.

B. ONE RING - ONE STRIP ELECTRODE CONFIGURATION

1. Questions

What is the magnitude of the electric field between the 3 and 5 V equipotentials (on centerline)? Indicate the direction on the graph paper.

$$2\text{V}/0.07\text{m} = 28\text{N/C}$$

What is the magnitude of the electric field between the 1 and 3 V equipotentials (on centerline)? Indicate the direction on the graph paper.

$$2\text{V}/0.08\text{m} = 25\text{N/C}$$

What is the magnitude of the electric field between the 7 and 9 V equipotentials (on centerline)? Indicate the direction on the graph paper.

$$2\text{V}/0.02\text{m} = 100\text{ N/C}$$

What is the magnitude of the electric field at coordinate (13,7)? Indicate the direction on the graph paper.

$$3.52-3.45/0.02 = 0.07/0.02 = 3.5$$

How can you describe the shape of the equipotentials at different regions of this electrode arrangement?

The shape of the equipotential at different regions of this arrangement are circular equipotential lines around the single ring on the side, all of the circular equipotential lines face the same direction. This is because there is only a single ring so the potential

Where is the E-field strongest (along the centerline)?

The E-field is the strongest when it is closest to the ring which is directed to the middle of the graph toward the concavity of the E-field.

Where is the E-field weakest (along the centerline)?

It is weakest when it is further away from the ring, the opposite side of the concavity of the E field.

What are the potential and the E-field inside the circular electrode?

On the left side, the potential is 10 volts, while on the other side it is zero volts, the E field for

both would be zero since conductors have a E field of zero.

COMPARING ELECTRODE CONFIGURATIONS

Compare the two configurations for which you mapped the electric fields. Discuss the similarities and differences between these electric field maps.

Both configurations formed rings like equipotential lines, but the distribution is different for both, while the two rings equipotential lines interact for the first lab this results in half of the equipotential lines facing one way while the other half face a different way. The electric field with a single ring form equipotential that faces single way along the entire field.